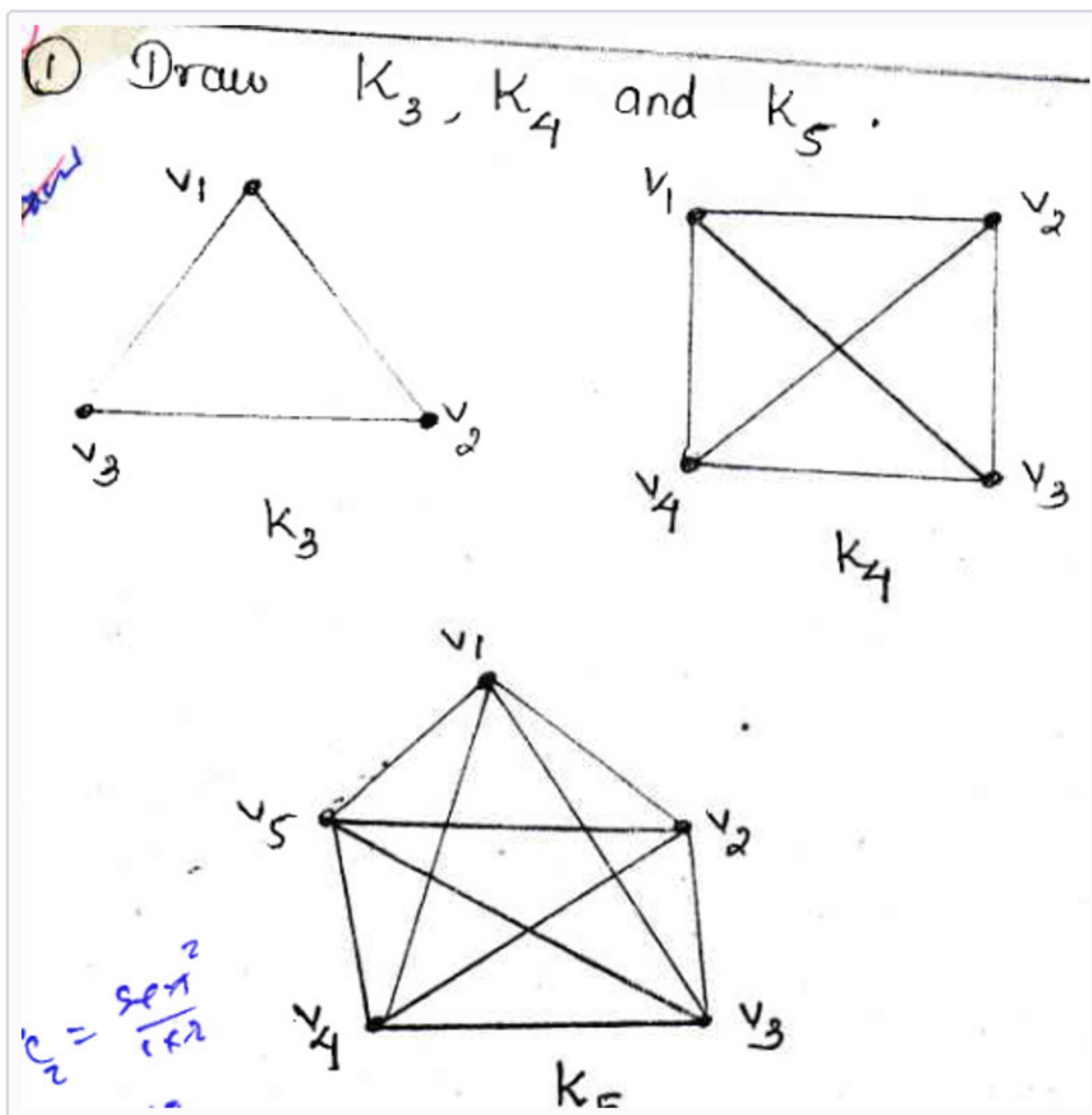


The supplied diagrams are included because the answers depend on reading the figure.

COMPLETE GRAPHS

1. Draw the complete graphs shown in the supplied figure:



2. Find a formula for the number of edges in the complete graph shown below.

$$|E(K_n)| = \binom{n}{2} = \frac{n(n-1)}{2}$$

A complete graph has one edge between every pair of distinct vertices.

The supplied diagrams are included because the answers depend on reading the figure.

COMPLETE BIPARTITE GRAPHS

3. Draw the complete bipartite graphs shown in the supplied figure:

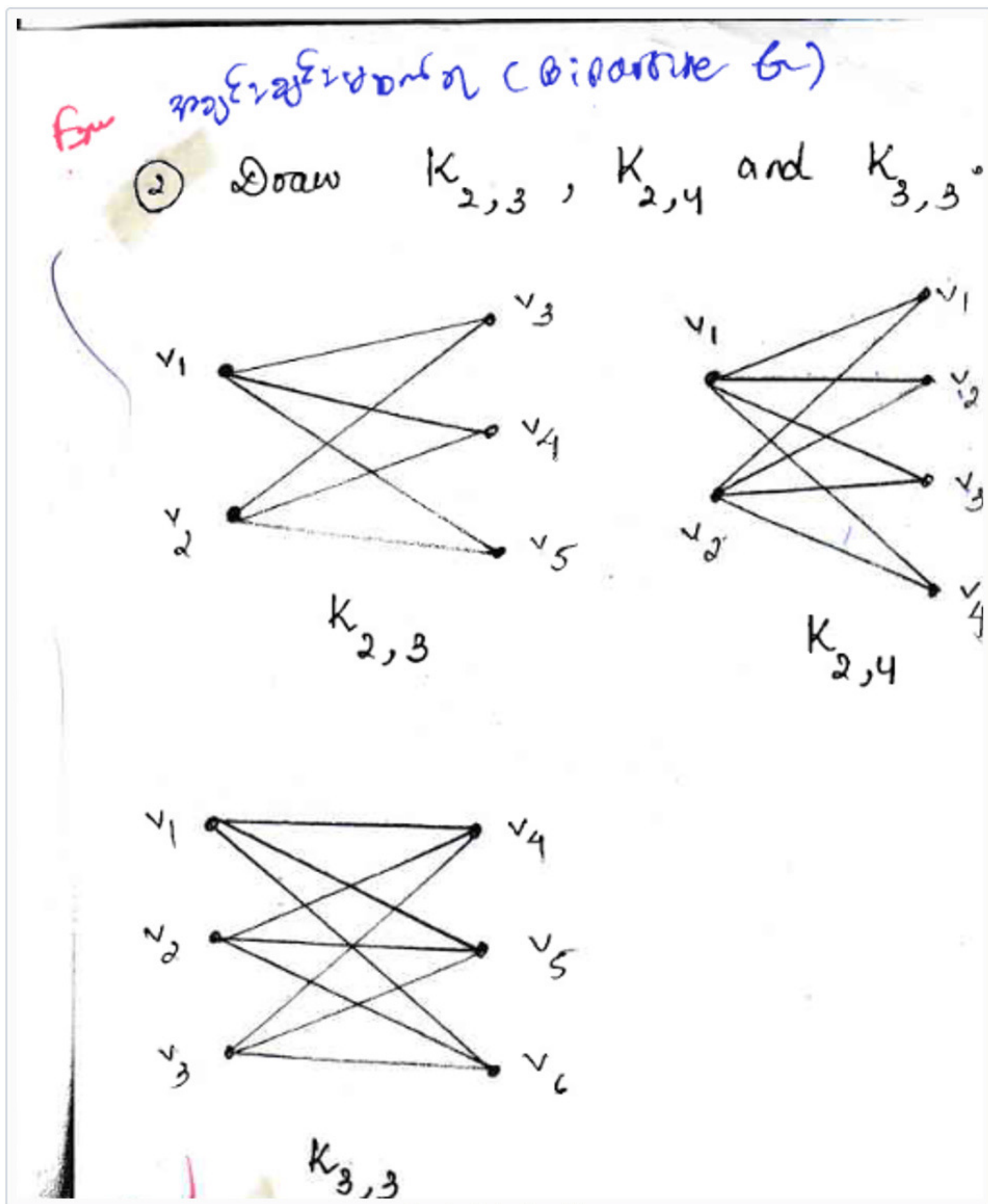


Figure for Q3: $K_{2,3}$, $K_{2,4}$ and $K_{3,3}$

Keep the two vertex sets separate; join every vertex in one set to every vertex in the other set.

The supplied diagrams are included because the answers depend on reading the figure.

BIPARTITE GRAPHS

4. For each supplied graph (a)–(d), if the graph is bipartite, specify the two disjoint vertex sets.

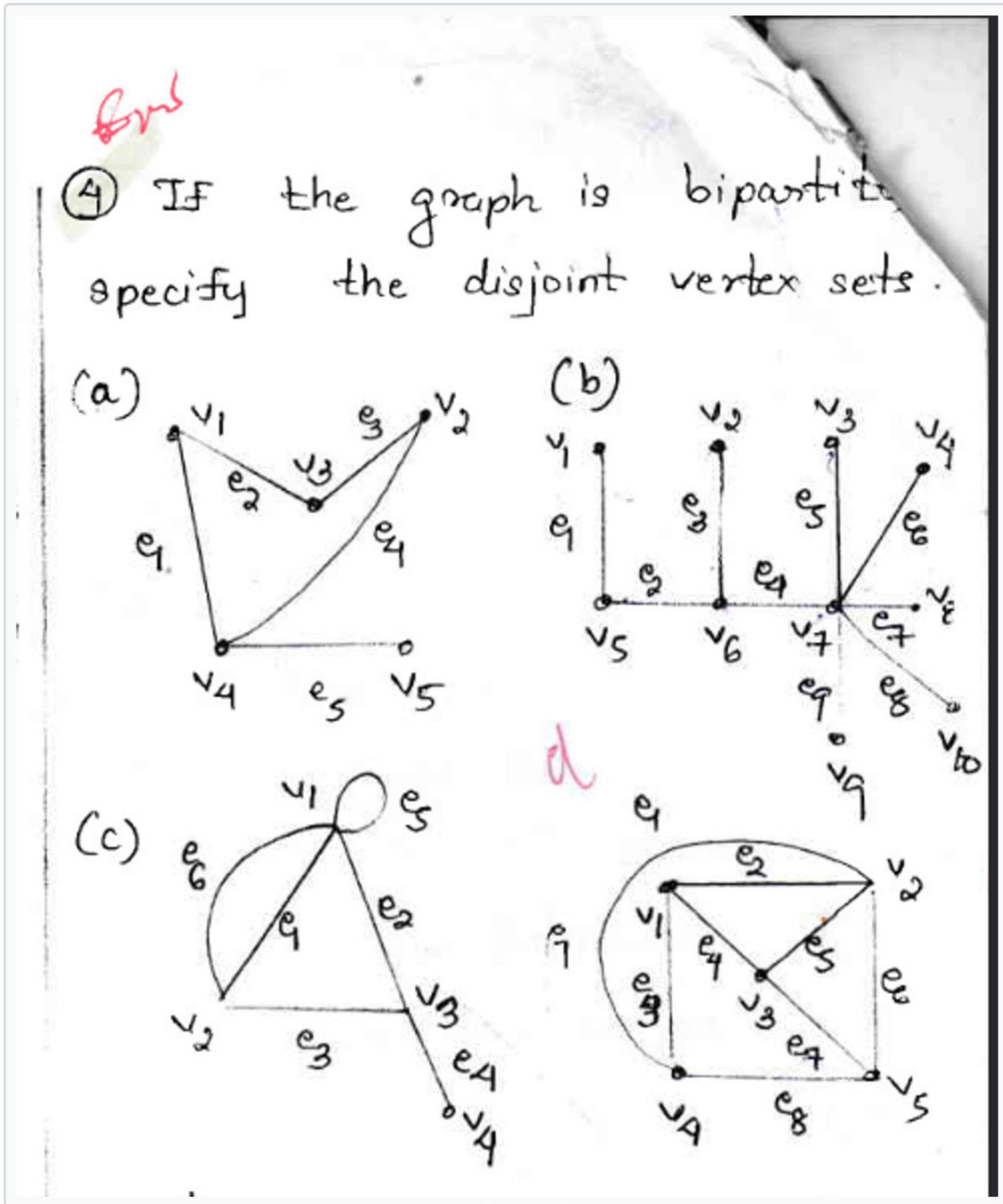


Figure for Q4: graphs (a)–(d)

5. Find a formula for the number of edges in the complete bipartite graph shown below.

$$|E(K_{m,n})| = mn$$

There are m vertices in one part and n vertices in the other part.

The supplied diagrams are included because the answers depend on reading the figure.

WEIGHTED PATHS

6. In the weighted graph shown, find a minimum-length path from v to w that passes through every vertex exactly once. Consider the three cases below:

$$v = a, w = e; \quad v = c, w = d; \quad v = a, w = b$$

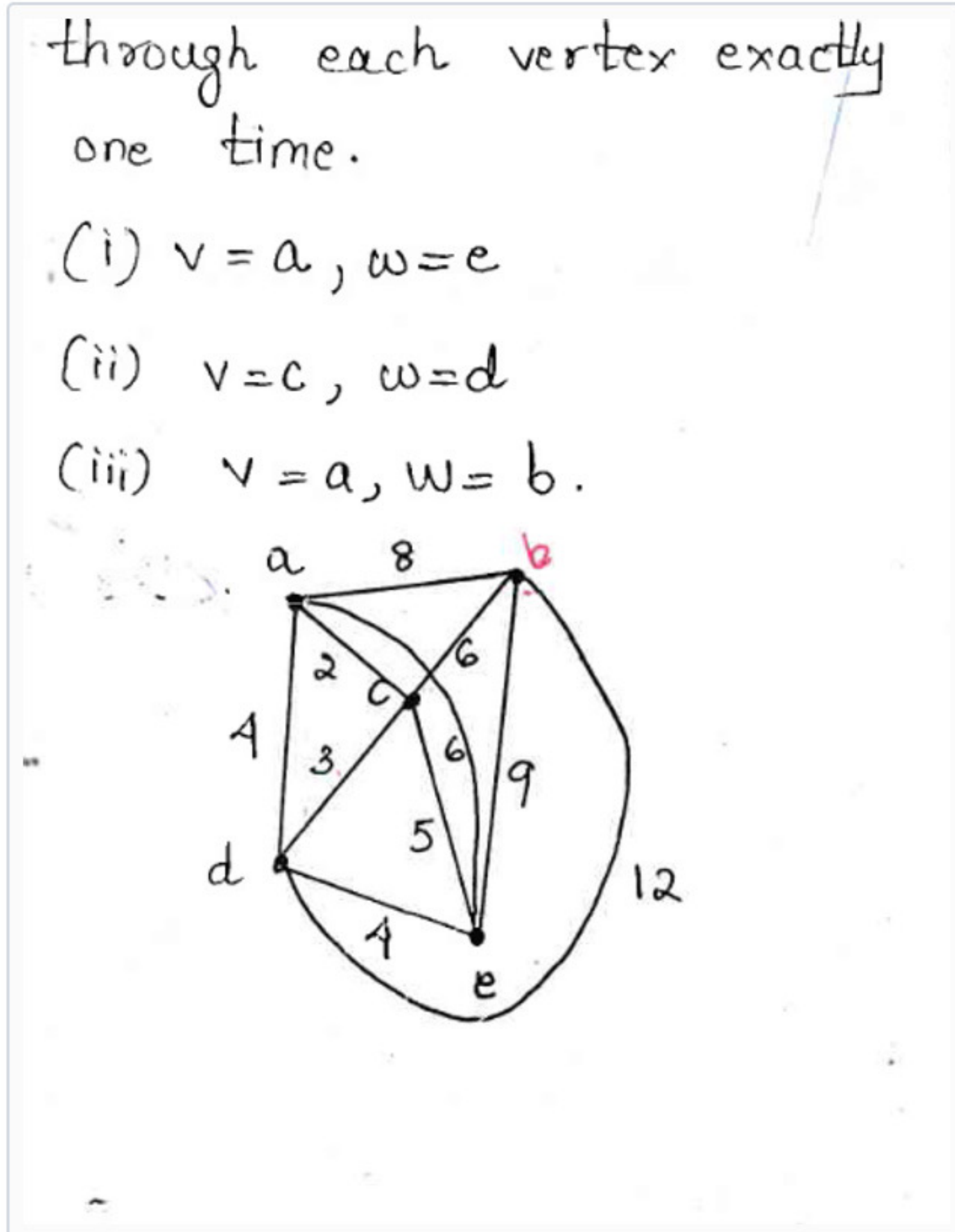


Figure for Q6: weighted graph

TREE AND ACYCLIC-GRAPH PROPERTIES

7. Draw a graph with the given properties, or explain why no such graph exists:
- (i) Tree; all vertices have degree 2.
 - (ii) Tree; six vertices have degrees 1,1,1,1,3,3.
 - (iii) Six edges and eight vertices.
 - (iv) Acyclic; four edges and six vertices.
 - (v) Tree; four internal vertices and six terminal vertices.

The supplied diagrams are included because the answers depend on reading the figure.

ROOTED TREE

8. Find the level of each vertex and the height of the rooted tree shown.

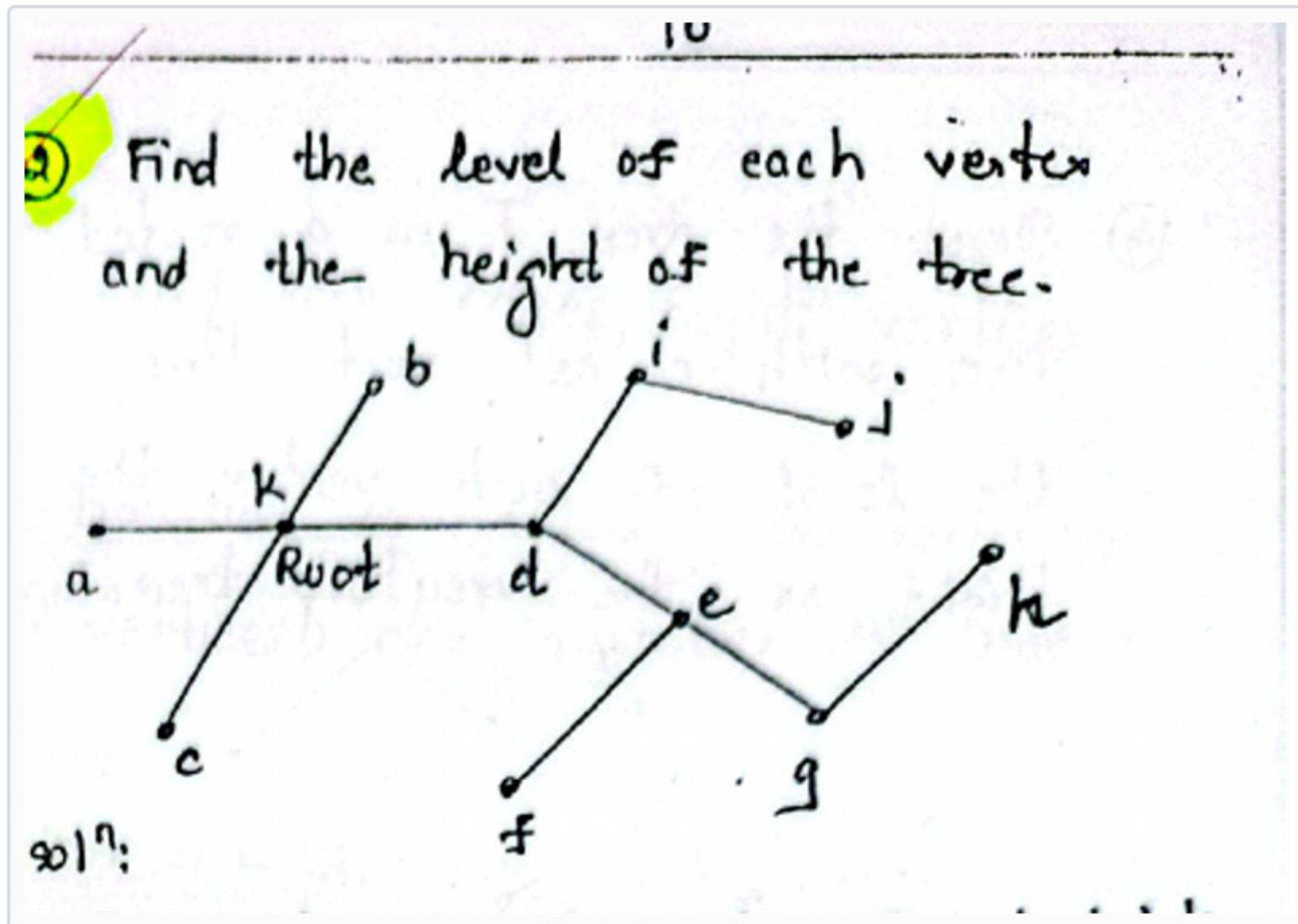


Figure for Q8: rooted tree

The supplied diagrams are included because the answers depend on reading the figure.

HUFFMAN CODING

9. Decode each bit string using the supplied Huffman code:
 (i) 01110100110 (ii) 0111100100110 (iii) 1110011101001111

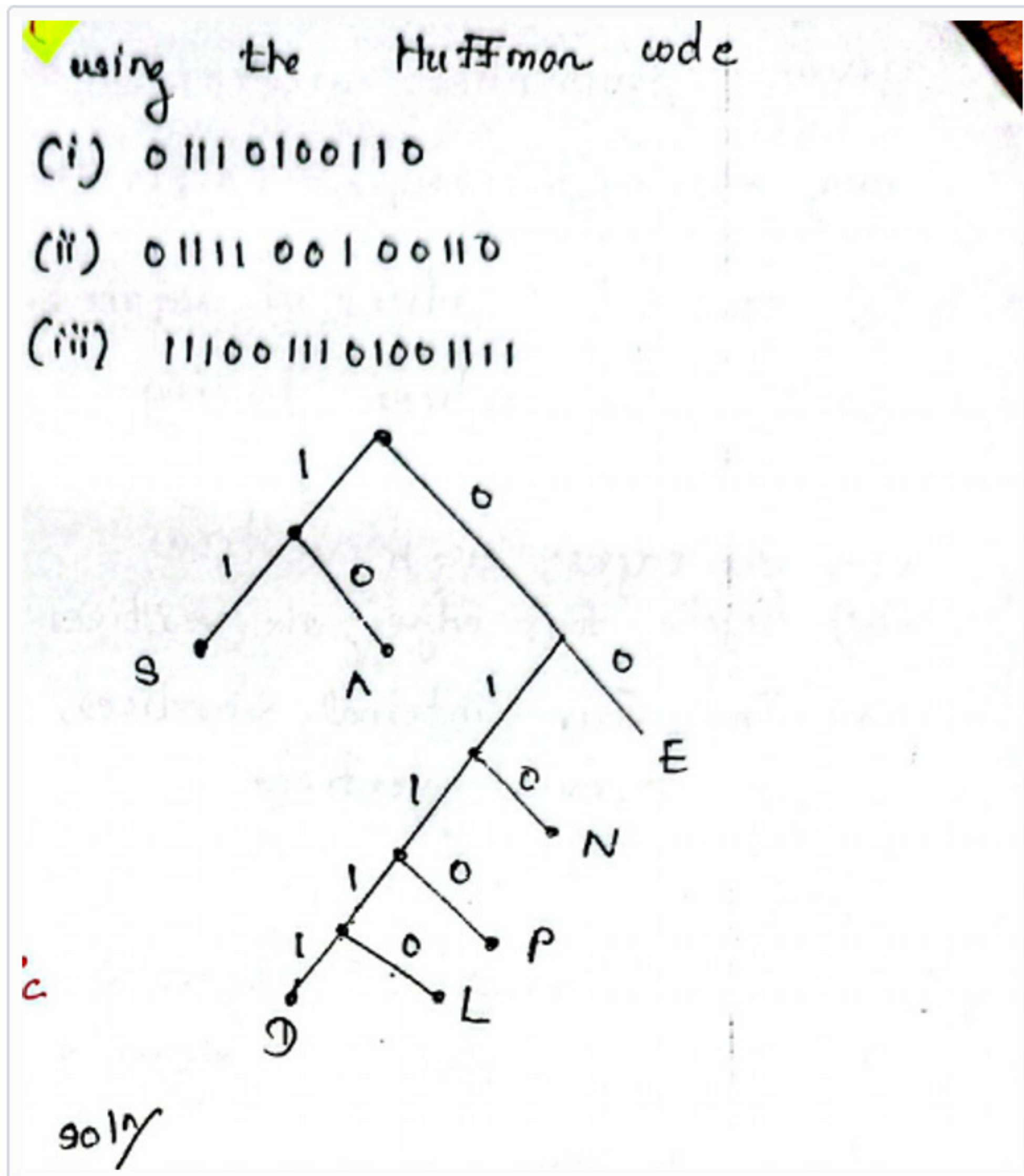


Figure for Q9: Huffman code tree

Follow the 0/1 branch labels from the root until a letter is reached.

The supplied diagrams are included because the answers depend on reading the figure.

EULER CYCLE

10. Determine whether the supplied graph G has an Euler cycle.
If it has one, exhibit an Euler cycle.

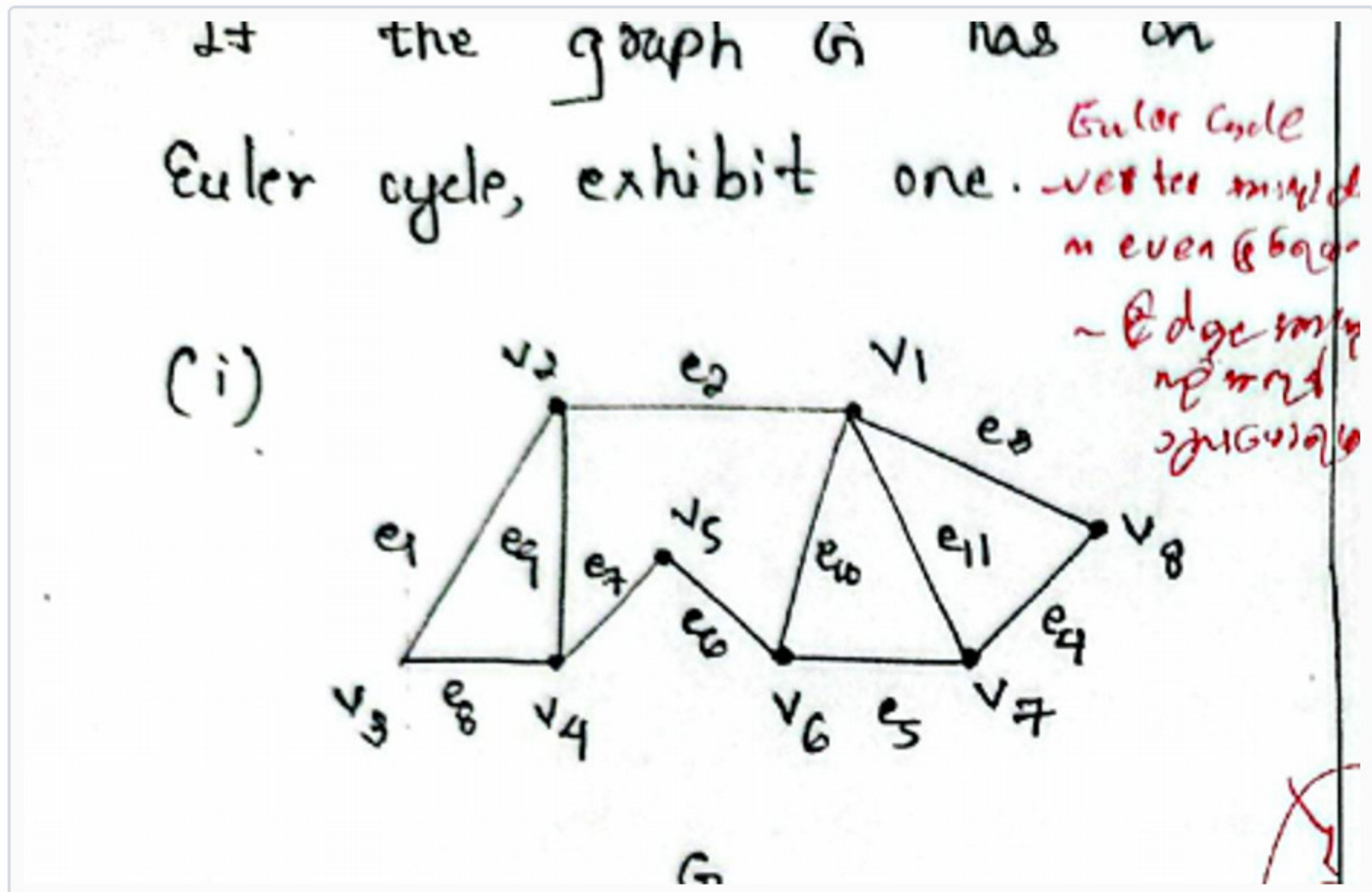


Figure for Q10: supplied graph G

Check the degree of every vertex first. A connected graph has an Euler cycle exactly when every vertex has even degree.